GENERAL SIR JOHN KOTELAWALA DEFENCE UNIVERSITY
FACULTY OF COMPUTING

INTAKE 41/42



Essentials of Artificial Intelligence (IT3182)

PROJECT REPORT

HYGEIA GLOVE - AI HEALTH MONITORING SYSTEM: AI-BASED INTELLIGENT HEALTH STATUS CLASSIFICATION AND EARLY RISK PREDICTION USING VITAL SIGNS

GROUP 04

Role	NAME	INDEX NUMBER
Project Manager	R. Dineth Meditha	D/COE/25/0011
Software Developer	K.H.Prabashwari Kavindi	D/BIT/24/0063
Data Analyst	H.K.K.Aberathna	D/BIS/24/0026
Quality Assurance Engineer	T.S Karunrathne	7165

Contents

Plagiarism Declaration 2
Abstract 3
1. Introduction 3
2. Problem Background 3
3. Problem Statement 4
4. Objectives 4
    4.1 Main Objective 4
    4.2 Specific Objectives 4
5. Selected AI Techniques 5
    5.1 Supervised Health Classification 5
    5.2 Feature Engineering 6
    5.3 RobustScaler 6
    5.4 Conv1D 6
    5.5 Bidirectional LSTM 6
    5.6 Temporal Monitoring 7
6. Justification of AI Techniques 7
7. System Design 8
8. Implementation Details 9
    8.1 Hardware Implementation 9
    8.2 ESP32 Firmware 10
    8.3 Dataset Preparation 10
    8.4 AI Model Implementation 11
    8.5 Backend Implementation 12
    8.6 Web Dashboard 12
9. Testing and Results 13
10. Limitations & Discussion 17
11. Ethical and Social Considerations 18
12. Future Enhancements 18
13. Conclusion 19
14. References 19
15. Individual Contribution Table 19

Plagiarism Declaration

We recognize that plagiarism is defined as taking someone else's ideas, work or information without giving it proper credit. We are aware of the significance of proper referencing, citing and ethical academic writing. We confirm we have followed the referencing guidelines and that our work will not be copied or presented by others as theirs.

NAME	INDEX NUMBER	SIGNATURE
R. Dineth Meditha	D/COE/25/0011	
K.H.Prabashwari Kavindi	D/BIT/24/0063	
H.K.K.Aberathna	D/BIS/24/0026	
T.S Karunrathne	7165	

Abstract

Continuous measurement of vital signs helps recognize any changes in the person's physiological state. Traditionally, there are many health monitoring methods which rely either on manually observed data or fixed threshold alerts. These techniques may recognize any deviations from the normal range; however, they have a limited number of features that could be analyzed at once. In this work, we introduce HYGEIA GLOVE, an AI-based intelligent health monitoring prototype, which aims to measure and process body temperature, blood oxygen saturation (SpO_2), and pulse rate. The hardware device is based on ESP32 microcontroller together with MAX30102 and MLX90614 sensors. Measurement data is sent via Wi-Fi to the Flask server, where the processing and health status classification are performed. For artificial intelligence, one deep learning network architecture was used, which combines Convolution1D and BiLSTM layers. Before feeding to the trained model, three raw vital signs and three features engineering variables are preprocessed by RobustScaler method. Health status is classified into three classes: Normal, Warning, and Critical. Web application shows actual measurements, predicted health status, and trend line. During testing phase, model showed the overall accuracy 93.98% and mean prediction confidence approximately 94.13%,10 Critical test samples. Critical-class evaluation was conducted on a small number of samples and requires further independent validation.

Keywords: Artificial Intelligence, Health Monitoring, HYGEIA GLOVE, Vital Signs, ESP32, Deep Learning, Conv1D, BiLSTM.

1. Introduction

Health monitoring is crucial for detecting any changes in a person's physiological state. Heart rate, blood oxygen saturation (SpO_2), body temperature are the metrics that are helpful in estimating the current state of a person. But the problem with traditional health monitoring is that it may require active participation of a person or the presence of predefined threshold values for some metrics. The idea of the original project was to develop an AI-based system that will analyze the vital signs, classify the current state of health and find potential health problems of a person. When the project was developed further, the idea was embodied in a physical prototype named HYGEIA GLOVE. HYGEIA GLOVE utilizes ESP32 microcontroller together with MAX30102 and MLX90614 sensors; MAX30102 sensor provides SpO_2 and pulse-rate metrics and MLX90614 sensor provides temperature measurements. The ESP32 sends collected data via Wi-Fi to the backend server where the AI-based classification decides whether the current state of health is in the Normal, Warning or Critical categories. The web dashboard provides current vital signs, classification and trend of the data. Thus, the project is about the integration of Artificial Intelligence, IoT sensors, web technologies and data processing in one prototype.

2. Problem Background

Vital signs like body temperature, SpO_2, and pulse rate offer critical insights into the physiological state of an individual. The conventional method of monitoring such vital parameters usually involves either manual monitoring or some kind of threshold value without

fully taking into account the interdependence among different vital parameters. Hence, HYGEIA GLOVE, which is a smart glove prototype based on the ESP32, has been designed to measure these vital parameters and send the data to a Flask server for AI-based processing and classification.

3. Problem Statement

Monitoring of vital signs is essential in assessing any change in a person's physiological state. Traditional systems of monitoring usually rely on manual observations or thresholds of alarms. Even though these approaches are helpful in identifying clearly abnormal values, they fail to consider several indicators or show the progression of the health status.

This problem can be solved through the development of the HYGEIA GLOVE system – a smart glove using ESP32 board that monitors the user's body temperature, SpO_2, and pulse rate values. The values collected via the glove are transferred through Wi-Fi to a Flask server where the values are classified into Normal, Warning, or Critical according to an AI model.

In addition to that, the project features a dashboard showing the current vital signs and 30 minutes trend. The objective of the system is to act as an intelligent health monitoring and decision-making prototype and not substitute medical professionals' work.

4. Objectives

4.1 Main Objective

To design and develop an AI-based intelligent system for analyzing vital-sign data to classify a person's current health status and predict early health risks using machine learning and intelligent decision-making techniques.

4.2 Specific Objectives

- **Smart Glove Development**

  To build a smart glove based on the ESP32 for measuring physiological signals.

- **Body Temperature Measurement**

  To measure the body temperature with the help of the MLX90614 sensor.

- **SpO_2 and Heart Rate Measurement**

  To collect the measurements of SpO_2 and heart rate with the MAX30102 sensor.

- **Wireless Data Transmission**

  To send sensor data from the ESP32 to the backend via Wi-Fi.

- **Data Pre-processing and Scaling**

To pre-process and scale collected physiological measurements before processing by AI algorithms.

- **Feature Engineering**

  To engineer features that will represent the relationships between different physiological signals.

- **Health Status Classification**

  To build a supervised deep learning model that can classify the health state as Normal, Warning, or Critical.

- **Flask REST API Development**

  To build a Flask REST API for communication between hardware, AI, and the web application.

- **Temporal Monitoring**

  To store recent measurements for monitoring over time.

- **Web Dashboard Development**

  To build a web dashboard displaying vital signals, health classification results, and trends.

- **AI Model Validation**

  To validate the performance of the AI model by using classification metrics.

- **Ethical Considerations**

  To address issues of privacy, reliability, bias, accuracy, and responsible use of AI in the health monitoring setting.

5. Selected AI Techniques

This final implementation relies heavily on supervised deep learning algorithms with feature engineering, normalization, and time-based monitoring.

5.1 Supervised Health Classification

The primary AI algorithm that will be used is multiclass classification. The algorithm will take in vital sign inputs and classify them into one of three categories:

Normal → Warning → Critical

There are three raw features extracted from the monitor:

- Temperature
- SpO_2
- Pulse rate

And three more engineered features:

- Temperature/SpO_2 ratio
- Pulse/SpO_2 ratio
- Health index

Thus, the classification algorithm takes in six input features.

5.2 Feature Engineering

Feature engineering helps to provide additional information to the AI model about the relationships between the individual vital signs. In addition to the sensor measurements, the system calculates:

Ratio Temperature/SpO_2 – It indicates the relationship between the body temperature and oxygen saturation.

Ratio Pulse/SpO_2 – It indicates the relationship between the pulse rate and oxygen saturation.

Health Index – It represents additional derived information on the physiological measurements of a person.

The idea is to give the model both initial and derived information.

5.3 RobustScaler

The input features may have different numerical ranges. For example, temperatures, SpO_2 in percents, pulses, and the calculated ratios have different scales.

The RobustScaler is applied to the features before they are supplied to the neural network.

5.4 Conv1D

The last model employs one-dimensional convolutional layers (Conv1D). The architecture includes Conv1D layers having:

- 64 filters
- 128 filters

Batch Normalization and Dropout are also used. Conv1D gives an opportunity to extract patterns from the input representation before the information will pass to the recurrent layers.

5.5 Bidirectional LSTM

The model includes Bidirectional Long Short-Term Memory (BiLSTM) layers as well. The implemented architecture includes:

- BiLSTM – 64 units
- BiLSTM – 32 units

Then the model employs dense layers including 128 and 64 units until reaching the output layer. The output layer employs Softmax with three outputs indicating Normal, Warning, and Critical.

5.6 Temporal Monitoring

The system tracks recent sensor values through the use of a rolling data buffer. The values are shown in the dashboard to allow the observation of temperature, SpO_2, and pulse rate changes over time. The implementation is done using the 30 minutes rolling history.

6. Justification of AI Techniques

The selected AI models have been picked depending on the requirements of the HYGEIA GLOVE system.

- **Supervised Classification**
  Supervised machine learning can be applied since it will help to classify input measurements to the predefined classes regarding health status:
- **Normal, Warning, and Critical.**
  Instead of presenting simple measurement results, the AI model will present its output in terms of categories which is more understandable to end-users.
- **Conv1D**
  Conv1D layer has been employed to learn the patterns based on the prepared feature representation. It serves as the feature-extraction layer before the recurrent part of the network.
- **Bidirectional LSTM**
  BiLSTM model has been selected as the component of the deep learning-based approach.
- **Feature Engineering**
  Feature engineering was chosen as it is possible that the health state depends not on the value of individual parameters but on relationships between them. Ratios and Health Index bring new data to the machine learning model.
- **RobustScaler**
  Scaling is necessary as the values of the six features are of different range. RobustScaler converts the features to a form suitable for processing by the neural network.
- **Temporal Monitoring**
  One value read by one sensor brings information for one moment. Keeping the values of recent readings helps users track whether they are changing or stable.

The combination of all these methods makes it possible for HYGEIA GLOVE to go beyond simple sensor readings and create intelligent monitoring prototype.

7. System Design

The HYGEIA GLOVE System Architecture involves five key layers, which include:

1. Sensor Layer

The smart glove is made up of:

- ESP32 Microcontroller
- MAX30102 Sensor
- MLX90614 Temperature Sensor

2. Communication Layer

The ESP32 links with the Wi-Fi and sends sensor measurements to the Flask server using HTTP requests.

3. Backend Processing Layer

The backend has been built using Python and Flask. The Flask application collects sensor measurements, preprocesses and prepares features, keeps track of the latest readings, and communicates with the AI model. Your progress report indicates that the Flask server, REST API, live data processing, and rolling data buffer have been created.

4. AI Classification Layer

The AI module has been built using Python. The Conv1D + BiLSTM model takes in six prepared features and makes prediction for one of three health-status categories:

- Normal
- Warning
- Critical

5. Presentation Layer

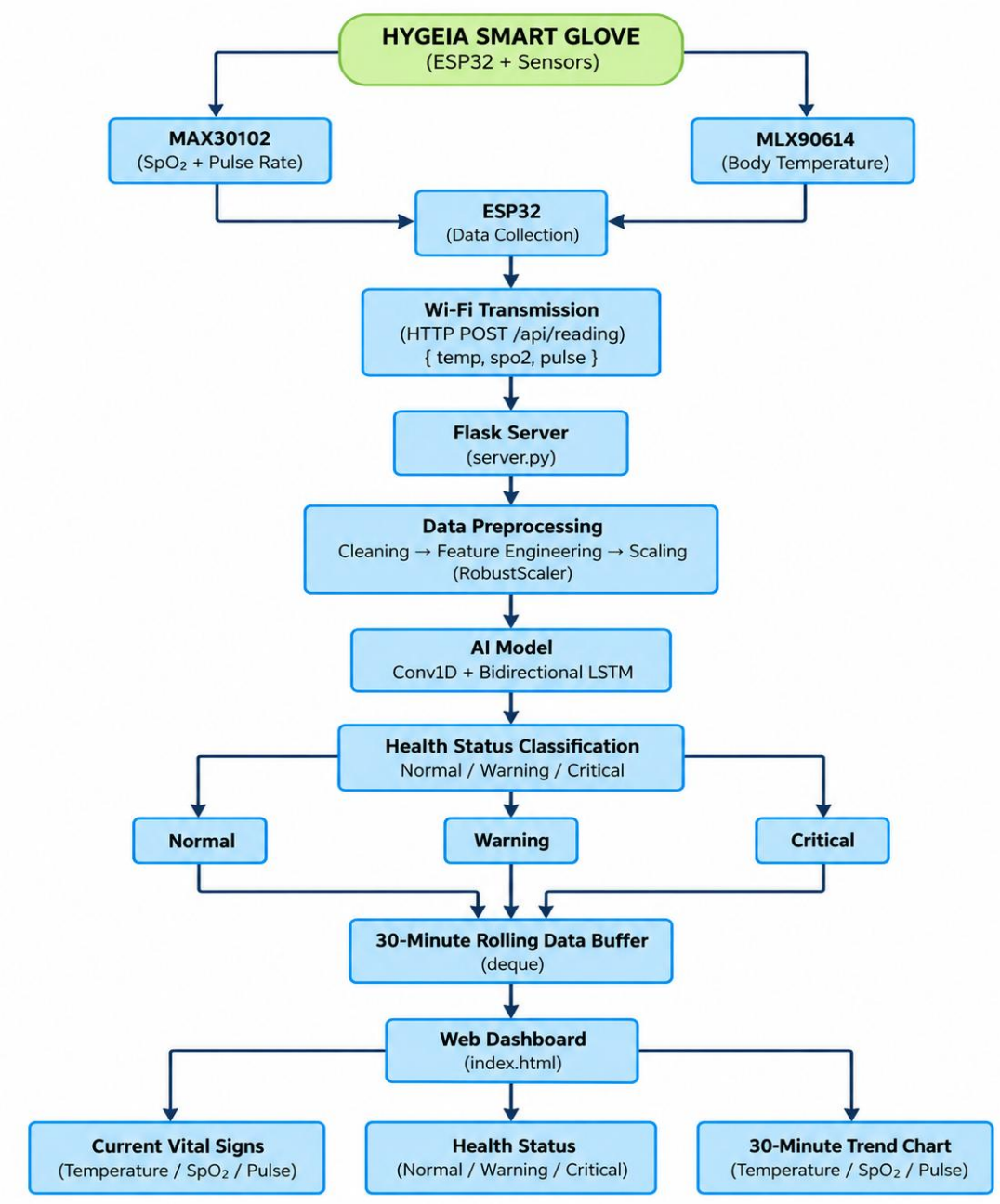
The web dashboard has been built using HTML, CSS, and JavaScript and integrated with the Python Flask backend. The following data are displayed:

- Latest temperature
- Latest SpO_2 level
- Latest pulse rate
- Health status prediction
- Trends in vital signs

The architecture of the system works as:

Sensors → ESP32 → Wi-Fi/HTTP → Python Flask Backend → AI Model → Web Dashboard

System Workflow



8. Implementation Details

8.1 Hardware Implementation

A physical prototype features an ESP32 as the microcontroller.
The following two sensors are attached to it:

MAX30102

The MAX30102 sensor is employed by the prototype for collecting the following data:

- SpO_2
- Pulse rate

MLX90614
This MLX90614 sensor measures the temperature of the human body.
These sensors send the data to the ESP32 microcontroller, which sends the data further to the computer running the Flask application.
The use of these sensors, implementation of Wi-Fi communication, and transmission of sensor data to Flask through HTTP POST requests.

8.2 ESP32 Firmware

The ESP32 firmware controls:

- Initialization of sensors
- Acquisition of sensor data
- Connection with Wi-Fi
- Preparation of sensor readings
- HTTP communications with server.

It sends data which includes:
Temperature, SpO_2 & Pulse Rate to the Flask backend.

8.3 Dataset Preparation

The following AI component were used for the vital-sign dataset stored :

data/patient_vitals.csv

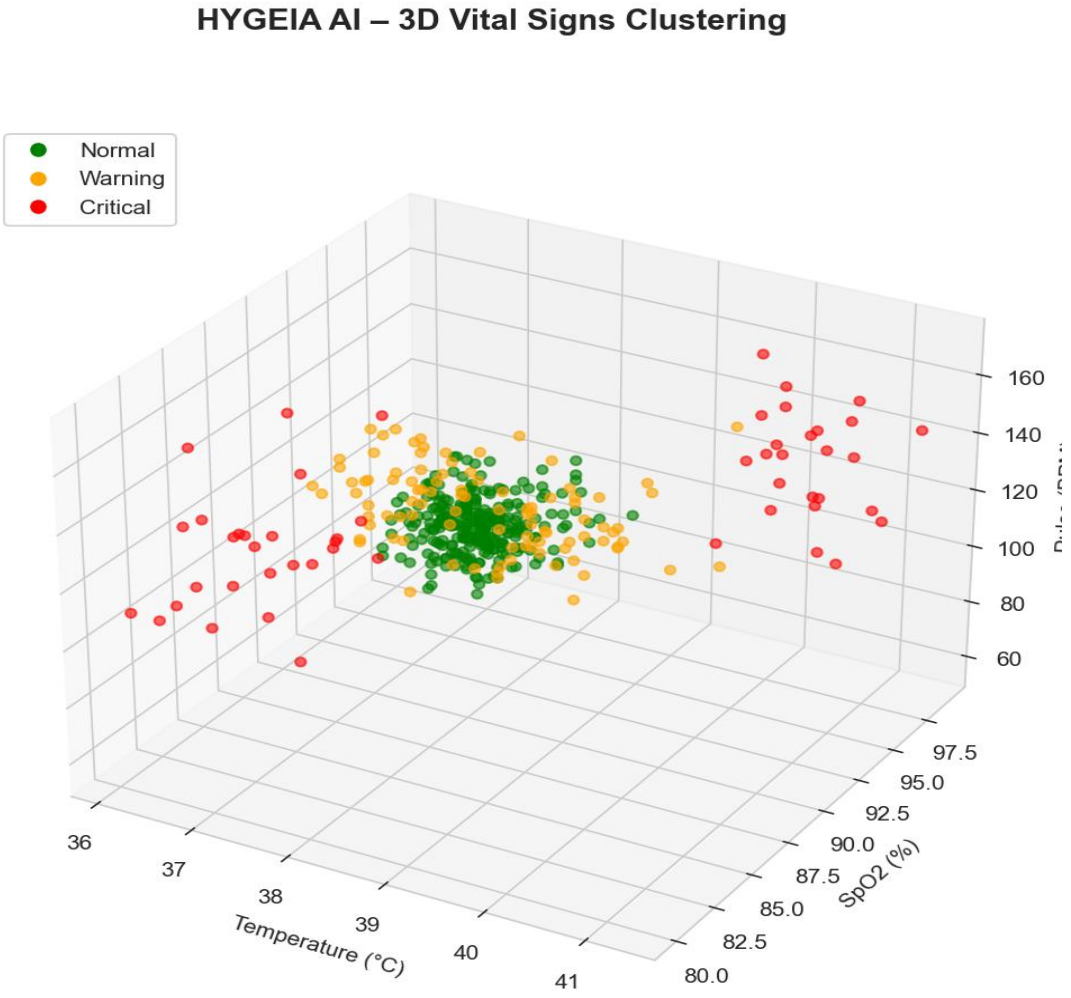
The important raw features are as follows:

Feature	**Description**
Temp	Body temperature
SpO2	Blood oxygen saturation
Pulse	Pulse rate (BPM)

Additionally, three engineered features were constructed

Engineered feature	**Purpose**
Temp_SpO2_ratio	Ratio of temperature to SpO2
Pulse_SpO2_ratio	Ratio of pulse to SpO2
Health_Index	Index reflecting the state of the person based on vital signs

The six features are scaled using RobustScaler and organized into sliding-window sequences of five consecutive readings, producing an input shape of (5, 6).



8.4 AI Model Implementation

The architecture makes use of the combination of Conv1D (Convolutional Neural Network) layers and BiLSTM (Bidirectional Long Short-Term Memory) layers.

Layer	**Configuration**
Input	Temp, SpO_2, Pulse + other engineered features
Conv1D	64 filters, kernel size 1
Batch Normalization	Applied
Dropout	0.3
Conv1D	128 filters, kernel size 1
Batch Normalization	Applied
Dropout	0.3

BiLSTM	64 units
Dropout	0.3
BiLSTM	32 units
Dropout	0.3
Dense	128 units, ReLU
Batch Normalization	Applied
Dropout	0.4
Dense	64 units, ReLU
Dropout	0.3
Output	Softmax of 3 classes

The final Softmax layer predicts the probability for Normal / Warning / Critical

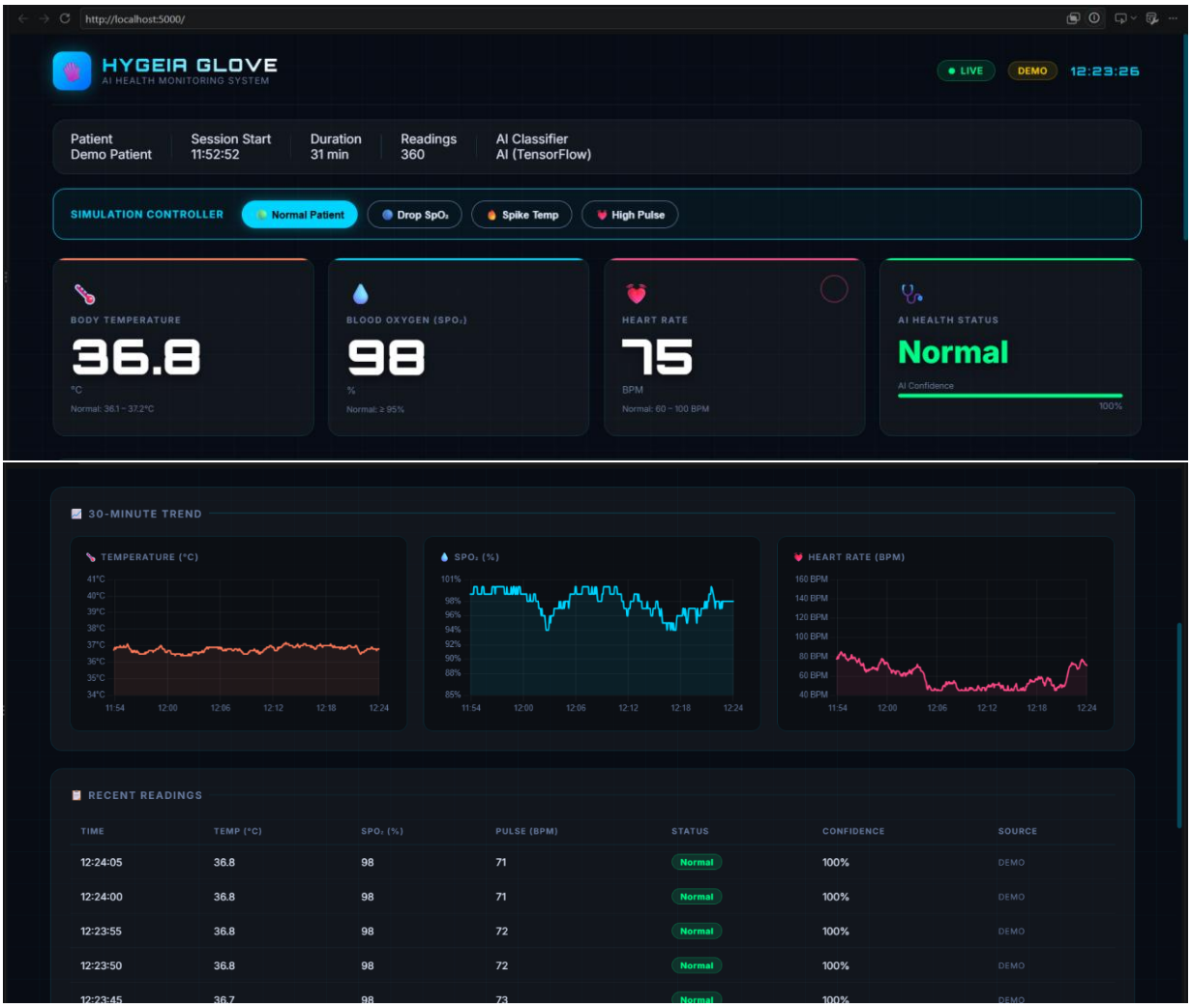
8.5 Backend Implementation

Flask backend has been utilized to link the hardware, AI, and the frontend.
RESTful endpoints have been created to receive and obtain health monitoring data.

Method	**Endpoint**	**Description**
GET	/	Opens dashboard
POST	/api/reading	Receive ESP32 readings
GET	/api/data	Recent readings retrieval
GET	/api/current	Latest readings retrieval
GET	/api/status	Server status check
POST	/api/sample/start	Starting of simulation data
POST	/api/sample/stop	Stopping of simulation data
POST	/api/simulation/mode	Changes simulation scenario
POST	/api/clear	Deletes current buffer

8.6 Web Dashboard

- A web dashboard was created utilizing HTML, CSS, JavaScript, and Chart.js together with the Flask backend.
- The web dashboard shows current data on temperature, SpO_2, heart rate, health status categorization by the AI (Normal, Warning, Critical), and the prediction confidence.
- A trend chart for 30 minutes demonstrates how vital parameters change. The web dashboard also contains controls to simulate various health conditions and shows which classification is used (AI or rules).



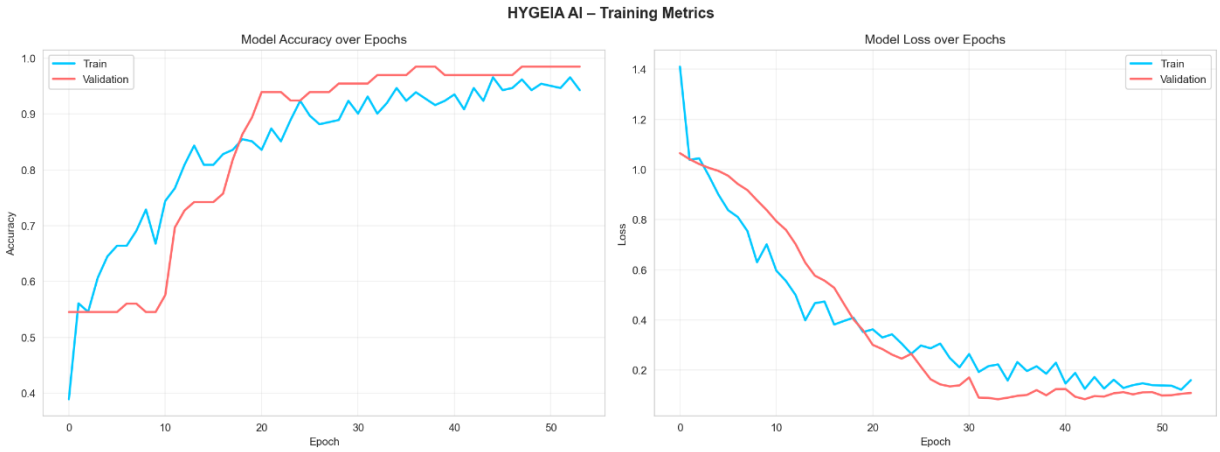
9. Testing and Results

Tests were conducted on the AI model and the prototype system as well.

AI Model Evaluation

The AI model was tested using 83 test data samples.

- Total Test Samples: 83
- Number of Correctly Classified Samples: 78
- Number of Incorrectly Classified Samples: 5
- Tested Accuracy: 93.98%
- Average Confidence Level of Prediction: 94.13%



Classification Results

The classification performance of the Conv1D + BiLSTM model is summarized below.

Class	Precision	Recall	F1 Score	Samples
Normal	0.98	0.96	0.97	53
Warning	0.89	0.85	0.87	20
Critical	0.83	1.00	0.91	10

Code Blame 19 lines (14 loc) · 650 Bytes Raw

```
HYGEIA AI – Patient Health Monitoring System
============================================================

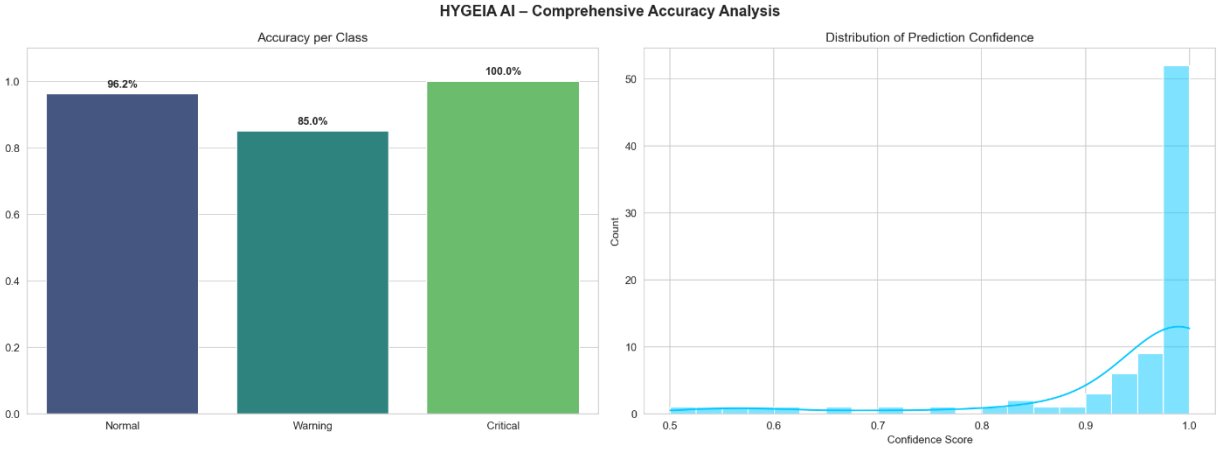
CLASSIFICATION REPORT
------------------------------------------------------------

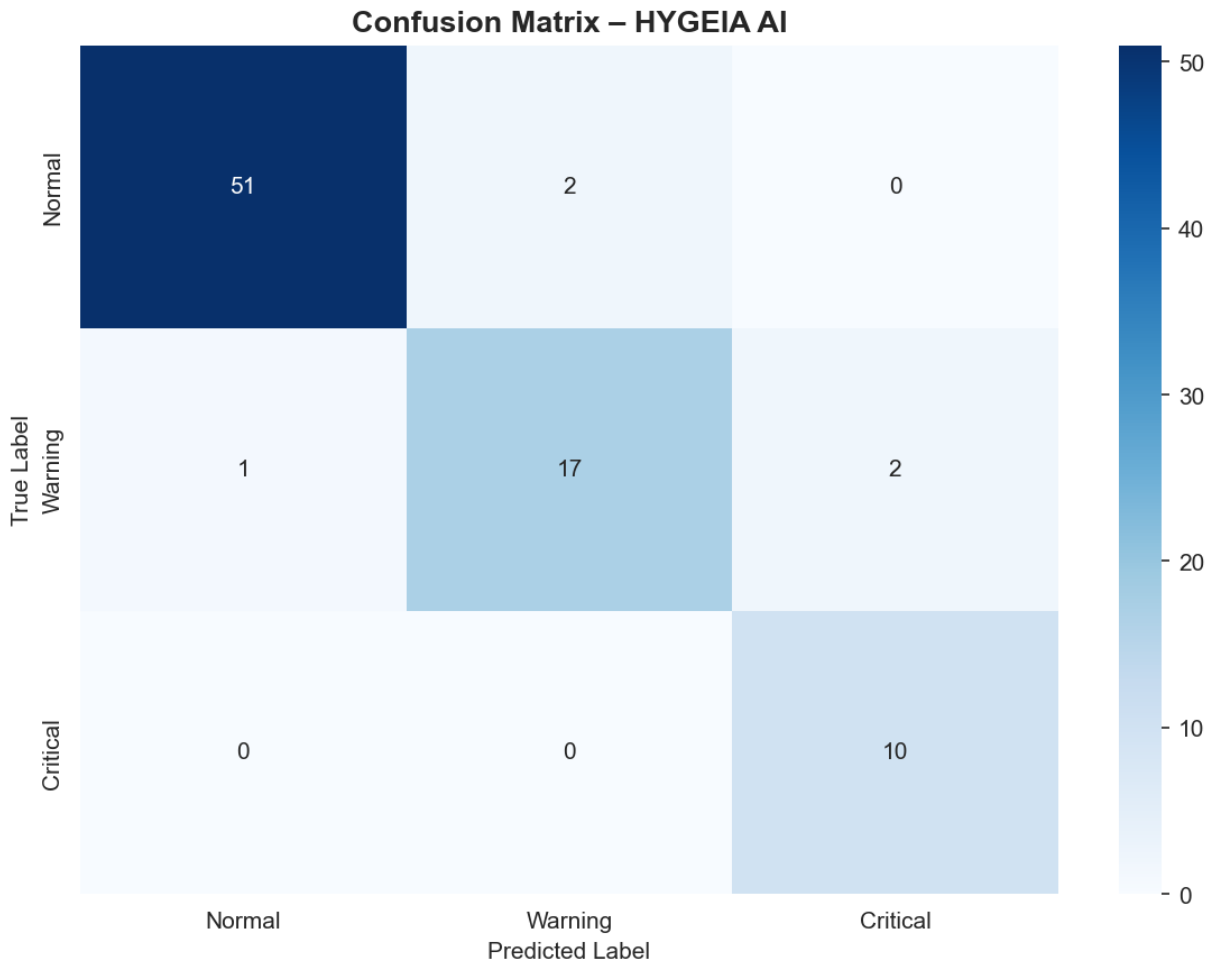
              precision    recall  f1-score   support

      Normal       0.98      0.96      0.97        53
     Warning       0.89      0.85      0.87        20
    Critical       0.83      1.00      0.91        10

    accuracy                           0.94        83
   macro avg       0.90      0.94      0.92        83
weighted avg       0.94      0.94      0.94        83

Overall Accuracy : 93.98%
Mean Confidence  : 0.9413
Confidence Std   : 0.1162
```



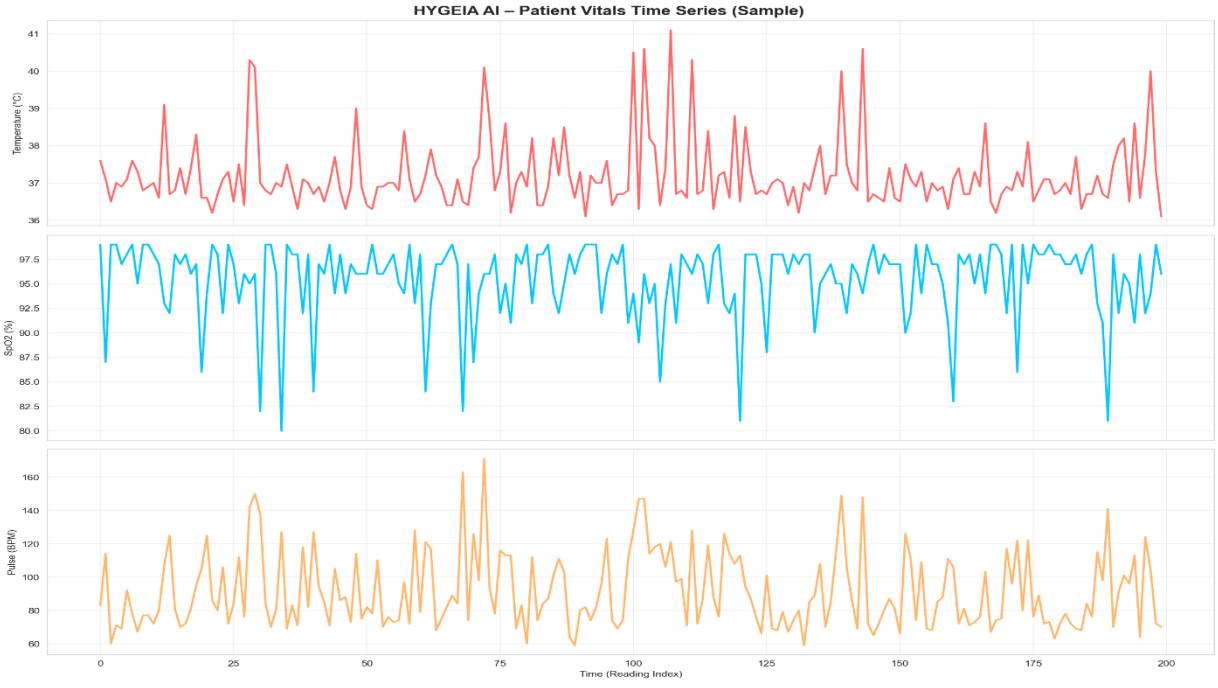


Hardware and Communication Testing

The implementation also confirmed that:

- The ESP32 microcontroller and the firmware.
- The MAX30102 sensor integration for SpO2 and heart rate.
- The MLX90614 sensor integration for measuring temperature.
- The Wi-Fi communication and data transmission over HTTP protocol.
- The Flask backend which receives and processes the data from sensors.
- The real-time visual representation of vitals on web dashboard.




HYGEIA AI – Patient Vitals Time Series (Sample)
Temperature (°C)
SpO2 (%)
Pulse (BPM)
Time (Reading Index)

Overall Result

Hygeia Glove perfectly illustrates how ESP32 sensors, Wi-Fi, Flask, Deep Learning, REST APIs, and real-time web dashboards can be incorporated.

The tested model managed to get 93.98% accuracy and 94.13% mean prediction confidence. Although the tested model showed great potential in all three classes of Normal, Warning, and Critical, it is necessary to test it on larger independent datasets, especially those related to the Critical class.

10. Limitations & Discussion

Even though HYGEIA GLOVE shows the viability of an AI-powered smart health monitoring system, some limitations exist.

- **Limited Dataset:** The size of the dataset is rather small. It should be noted that the deep learning system usually requires a large and diverse dataset.
- Cri**tical-Class Limitation:** The test dataset consisted of only 10 Critical instances. Despite the promising results obtained from the model, more testing involving a greater variety of Critical instances is needed.
- **Sensor Noise:** The sensor measurement could be affected by the movement, position, environment, and noise. The problem of the finger movement and the sensor noise has been pointed out during the progress.
- **Temperature Calibration:** Temperature sensors could require the calibration for accurate measurements in different conditions.
- **Wi-Fi Connection:** The communication between the ESP32 and Flask backend is provided by the network connection. The unstable Wi-Fi connection could interrupt the real-time transmission.
- **Prototype, Not Medical Device:** The system hasn't been clinically validated and certified yet.
- **Generalization**: Limited training and test data imply that the system's performance for different users, ages, states, environments and sensor positions can not be estimated.
- **Model Evaluation:** The current training pipeline leads to overlapping sequences, where scaling is performed before the split of the test dataset, resulting in potential data leakage. It is essential to have independent testing to validate the results in real-world settings.

11. Ethical and Social Considerations

- **Privacy:** Safeguard sensitive vital sign and personal health information from any breaches.
- **Data Security:** Ensure secure data transfer through communication and authentications during transmission of health data.
- **Bias in AI:** Small or biased samples may influence performance of models with respect to particular users.
- **Wrong Predictions:** Wrong predictions can be made, leading to wrong classifications of health statuses.
- **Human Responsibility:** The system shall assist users and healthcare professionals, but never substitute professional medical opinion.
- **Transparency:** It shall be understood by users that classifications of health are produced using AI and are prone to errors.
- **Responsible Use:** HYGEIA GLOVE is an educational prototype and is not medically certified or diagnosed tool.

12. Future Enhancements

While HYGEIA GLOVE presents a case study of AI-enabled health condition monitoring, there is scope for enhancing reliability, usability, and clinical relevance of this system.

- Automated Alerts: Provide SMS, email or mobile alerts in cases where there may be serious health conditions.
- Improved Data and Model Verification: Obtain additional representative physiological measurements, especially of the Critical class, and validate the model using independent testing data.
- Cloud Hosting: Host the backend in a secure cloud environment for remote monitoring capabilities.
- Patient Management: Implement patient profiles along with individual monitoring records.
- Mobile Application: Build an application interface that would facilitate access to vital sign readings.
- Sensor Improvement: Improve calibration, noise filtering, and noise handling of sensors for more accurate measurements.

13. Conclusion

The HYGEIA GLOVE – AI Health Monitoring System consists of AI, Internet of Things, embedded devices, and web applications, providing intelligent health monitoring services. This smart glove measures the temperature, SpO_2, and pulse rate and transmits them to the Flask server, where the information is further processed and classified using AI models. The model classifies the health state into Normal, Warning, and Critical, while the dashboard provides an insight into the actual vital signs and their trends. The model reaches 93.98% accuracy and around 94.13% mean prediction confidence. Nevertheless, The Critical class was evaluated using 10 samples; further independent validation is required.

14. References

1. Breiman, L. (2001). Random forests. *Machine Learning, 45*, 5–32. https://doi.org/10.1023/A:1010933404324
2. Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation, 9*(8), 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735
3. Zadeh, L. A. (1965). Fuzzy sets. *Information and Control, 8*(3), 338–353. https://doi.org/10.1016/S0019-9958(65)90241-X
4. World Health Organization. (2021). *Ethics and governance of artificial intelligence for health: WHO guidance*. World Health Organization. https://www.who.int/publications/i/item/9789240029200
5. Russell, S., & Norvig, P. (2021). *Artificial intelligence: A modern approach* (4th ed.). Pearson.

15. Individual Contribution Table

Member	Main Area	Individual Contribution
R. Dineth Meditha	ML Classification & AI Model Development	Dataset preprocessing, feature engineering, AI model development, model training, testing, evaluation and performance analysis
K.H. Prabashwari Kavindi	AI Integration, Frontend & Dashboard	AI prediction integration, dashboard development, graphs, health-status visualization, alerts and frontend testing
H.K.K. Aberathna	Backend & Integration	Flask REST API development, real-time sensor-data processing, backend implementation and system integration
T.S. Karunrathne	Risk Prediction & Trend Analysis	Risk classification, vital-sign trend analysis, health-status monitoring (Normal, Warning, Critical), and testing of risk-monitoring functionality.